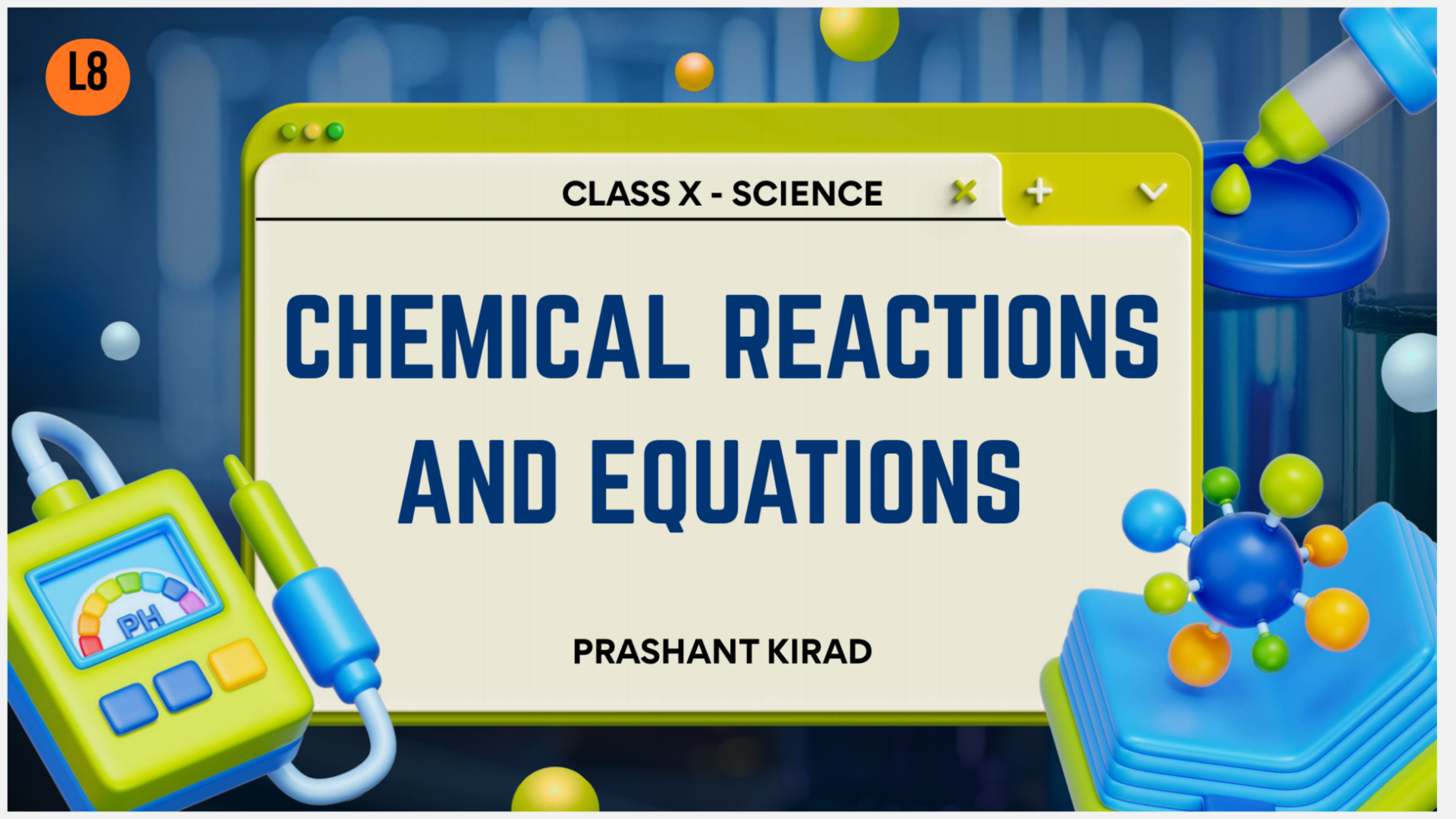


L8

CLASS X - SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

PRASHANT KIRAD



Corrosion

Corrosion is the attack on a metal due to chemical reactions with substances like moisture, acids, and oxygen present in the surrounding environment.

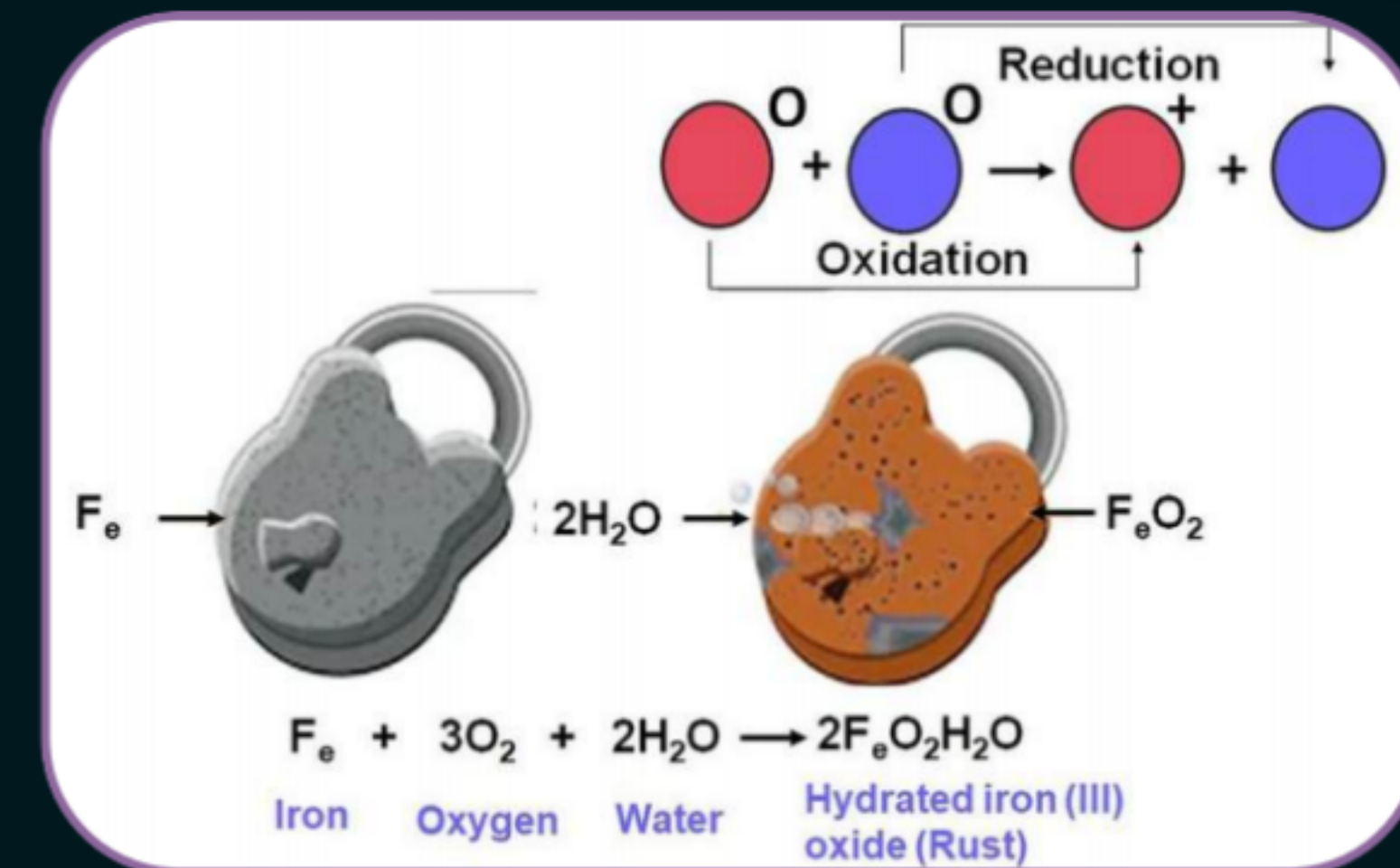
In simple terms: Destruction of metal due to reaction with air and moisture is called corrosion.



Examples of Corrosion

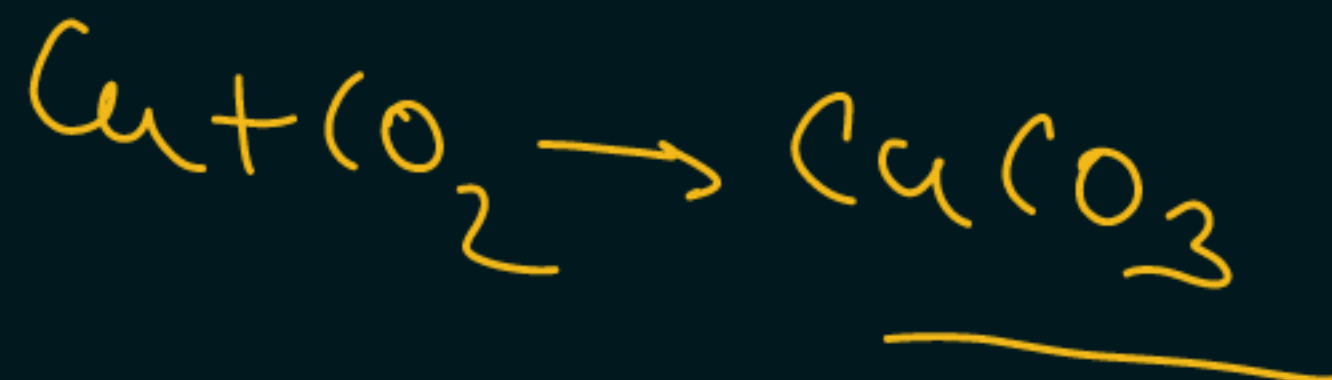
1. Rusting of Iron

- Fresh iron is shiny but turns reddish-brown after some time.
- This is due to the formation of iron oxide (rust).
- (Iron + Oxygen + Water → Hydrated Iron Oxide)



2. Black Coating on Silver

- Silver reacts with hydrogen sulphide in air to form silver sulphide (Ag_2S).
- This forms a black coating.



3. Green Coating on Copper

- Copper reacts with carbon dioxide, oxygen, and moisture in air.
- Forms basic copper carbonate which appears green.
- Example: Statue of Liberty turned green due to corrosion.



Harmful Effects of Corrosion

- *Damages iron structures like cars, bridges, railings, and pipelines.*
- *Causes weakening of materials.*
- *Leads to heavy financial losses for repair and replacement.*

Preventive Measures

- *Painting*
- *Galvanization*
- *Oiling and greasing*
- *Coating with non-reactive metals*

LET'S TEST YOUR MEMORY



Q. Why do we apply paint on iron articles?
(NCERT)

Ans -

We apply paint on iron articles to prevent rusting.

Paint forms a protective layer that prevents iron from coming into contact with air (oxygen) and moisture, which are necessary for oxidation (rusting) to occur.

Rancidity

Rancidity is the condition produced by the oxidation of fats and oils in food materials, which results in a bad taste and unpleasant smell.

Examples: Ghee or butter kept open for a few days may start to smell foul, Fried foods like chips become stale and develop an unpleasant odor after prolonged exposure to air.



LET'S TEST YOUR MEMORY



Q. Rashmika wants to increase the shelf life of fried snacks without using chemical preservatives. She is considering the following three changes in packaging:

Method (I): Packing chips in airtight transparent packets.

Method (II): Packing chips in airtight packets filled with nitrogen gas.

Method (III): Packing chips in vacuumsealed packets and storing them away from sunlight.

(i) Which method would be the most effective in preventing rancidity?

(ii) Give a scientific reason for your choice based on oxidation

LET'S TEST YOUR MEMORY



Ans -

- (i) Method (II) – Packing chips in airtight packets filled with nitrogen gas.**
- (ii) Nitrogen is an inert gas and does not react with oils and fats. It replaces oxygen inside the packet, thus preventing oxidation (rancidity). Without oxygen, the oxidation reaction cannot occur.**

LET'S TEST YOUR MEMORY



Q. A student in Rashmika's class argues that refrigeration alone is sufficient to prevent rancidity in fried food items. Do you agree with the student? Justify your answer using the concept of oxidation and reaction rate

Ans -

No, I do not agree.

Refrigeration only slows down the rate of oxidation by lowering temperature, but it does not stop it completely. Oxygen is still present, so oxidation (rancidity) can still occur, just more slowly.

To effectively prevent rancidity, oxygen must be removed or replaced (e.g., by nitrogen flushing) in addition to lowering temperature.

LET'S TEST YOUR MEMORY



Q. Oil and fat containing food items are flushed with nitrogen. Why? (NCERT)

Ans -

Oil and fat containing food items are flushed with nitrogen to prevent oxidation.

Oxygen in air can oxidize fats and oils, causing rancidity (bad smell and taste). Nitrogen, being an inert gas, replaces oxygen and thus prevents spoilage of the food.

LET'S TEST YOUR MEMORY



Q. A restaurant makes use of two types of cooking oils: Oil 'X' (rich in unsaturated fatty acids) and oil 'Y' (highly rich in saturated fatty acids). Both are kept in transparent containers near the kitchen window. After a few weeks, oil 'X' acquires a foul and unpleasant odour, whereas oil 'Y' does not.

(A) Identify the phenomenon affecting oil 'X' and explain why oil 'Y' is unaffected, referencing its chemical composition.

(B) The restaurant owner intends to keep oil 'X' in the refrigerator but sees condensation within the bottle. Predict the influence of moisture on the phenomenon discussed in (A) and propose an additional strategy to slow it down besides refrigeration.

LET'S TEST YOUR MEMORY



Ans -

(A) The phenomenon is rancidity (oxidation of oils).

Oil X (unsaturated) contains double bonds, which are more reactive and easily oxidised by air and light, producing a foul smell.

Oil Y (saturated) has no double bonds, so it is more stable and does not oxidise easily.

(B) Moisture (condensation) can accelerate rancidity by promoting oxidation and microbial activity.

To slow it down:

Store oil in airtight, dark containers (limit oxygen and light) or add antioxidants.

Chapter Complete...

